

minority of patients become compulsive gamblers. Since such gambling is associated with their seeing what they believe to be clear patterns in random numbers, this illustrates the *relation between knowledge and randomness*. It also shows that some aspects of what we call “knowledge” (and what I call narrative) are an ailment.

Once again, I warn the reader that I am not focusing on dopamine as the *reason* for our overinterpreting; rather, my point is that there is a physical and neural correlate to such operation and that our minds are largely victims of our physical embodiment. Our minds are like inmates, captive to our biology, unless we manage a cunning escape. It is the lack of our control of such inferences that I am stressing. Tomorrow, someone may discover another chemical or organic basis for our perception of patterns, or counter what I said about the left-brain interpreter by showing the role of a more complex structure; but it would not negate the idea that perception of causation has a biological foundation.

Andrey Nikolayevich’s Rule

There is another, even deeper reason for our inclination to narrate, and it is not psychological. It has to do with the effect of order on information storage and retrieval in any system, and it’s worth explaining here because of what I consider the central problems of probability and information theory.

The first problem is that information is *costly to obtain*.

The second problem is that information is also *costly to store*—like real estate in New York. The more orderly, less random, patterned, and *narratized* a series of words or symbols, the easier it is to store that series in one’s mind or jot it down in a book so your grandchildren can read it someday.

Finally, information is costly to manipulate and retrieve.

With so many brain cells—one hundred billion (and counting)—the attic is quite large, so the difficulties probably do not arise from storage-capacity limitations, but may be just indexing problems. Your conscious, or working, memory, the one you are using to read these lines and make sense of their meaning, is considerably smaller than the attic. Consider that your working memory has difficulty holding a mere phone number longer than seven digits. Change metaphors slightly and imagine that your consciousness is a desk in the Library of Congress: no matter how many books the library holds, and makes available for retrieval, the size of your desk sets some processing limitations. Compression is vital to the performance of conscious work.

Consider a collection of words glued together to constitute a 500-page book. If the words are purely random, picked up from the dictionary in a